DASC 3240, L07

Plotting two variables: Barplot and more

2023/24 SPRING SEMESTER

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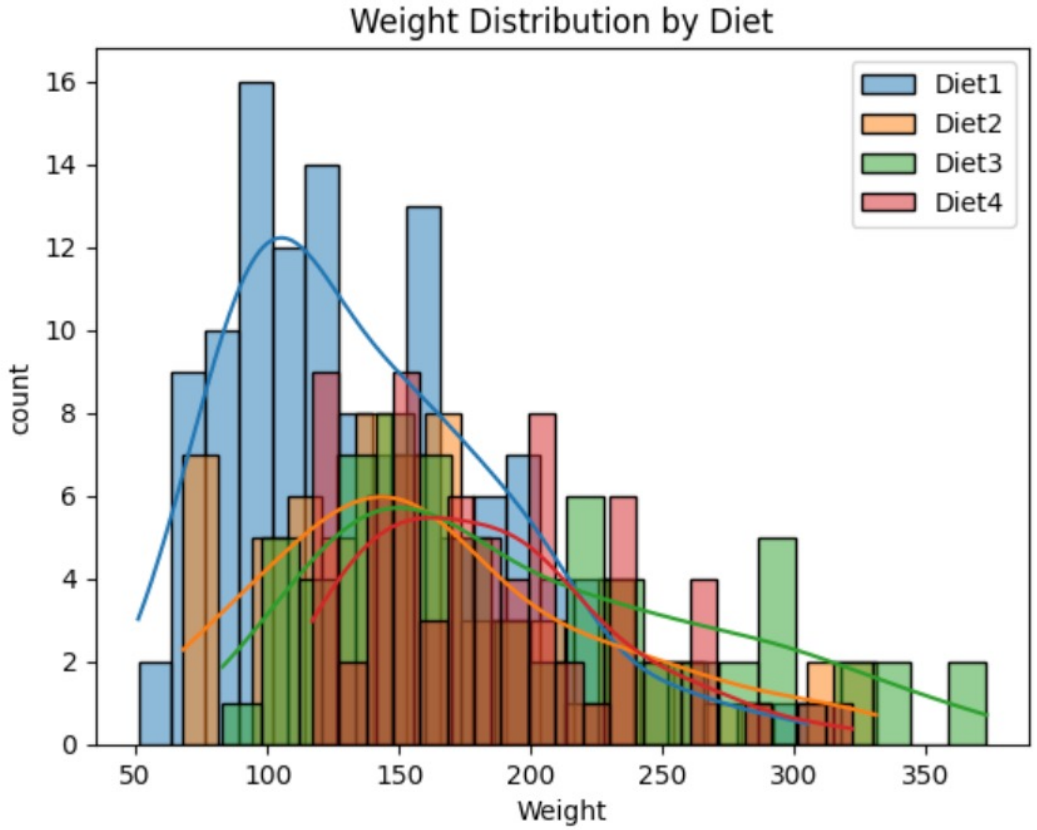
Notes from the previous assignment

- Unit
  - If values have unit, please describe it in a parenthesis. For example, ”**Weight (g)**”.
- Seaborn and matplotlib
  - Seaborn was developed based on matplotlib, so if “fig” and/or “ax” objects that were created by Seaborn can also be edited by matplotlib functions.

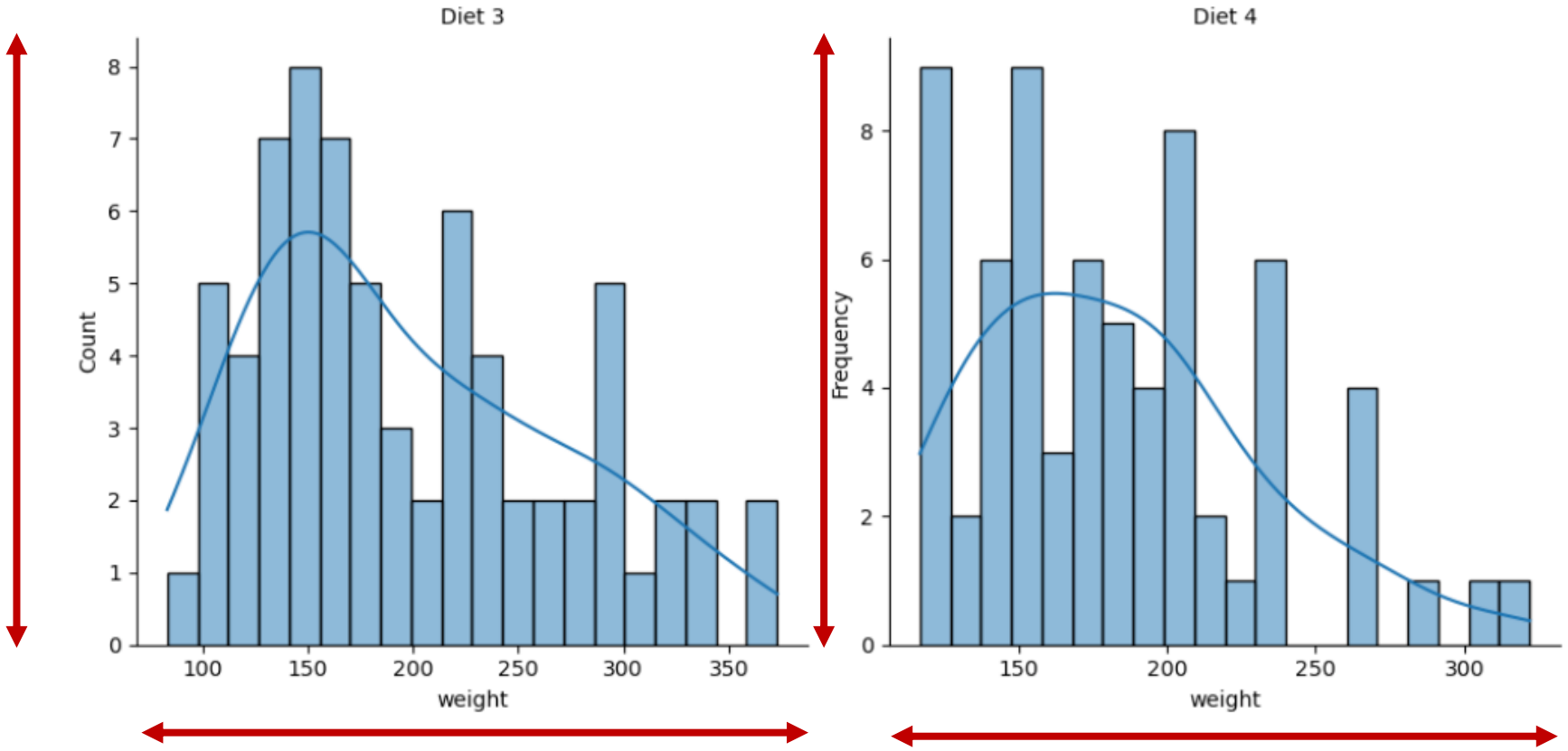
Notes from the previous assignment

- Bin size
  - If you overlay different histograms, you should use the same bin size for all the histograms.



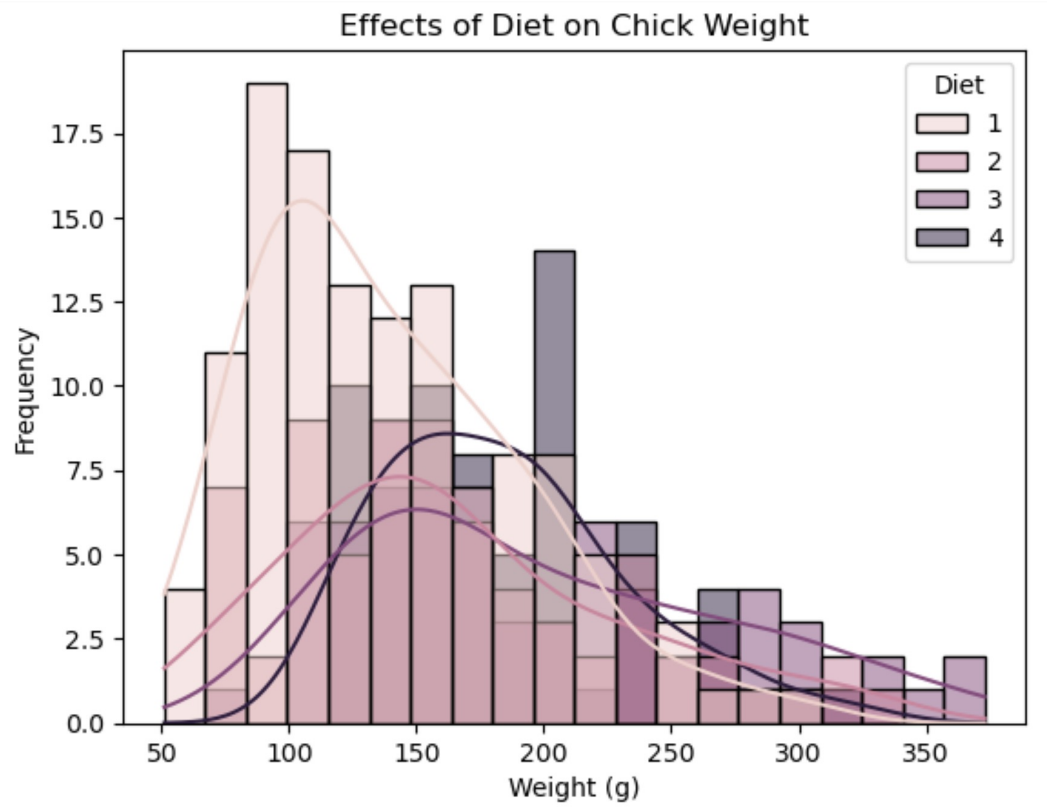
Notes from the previous assignment

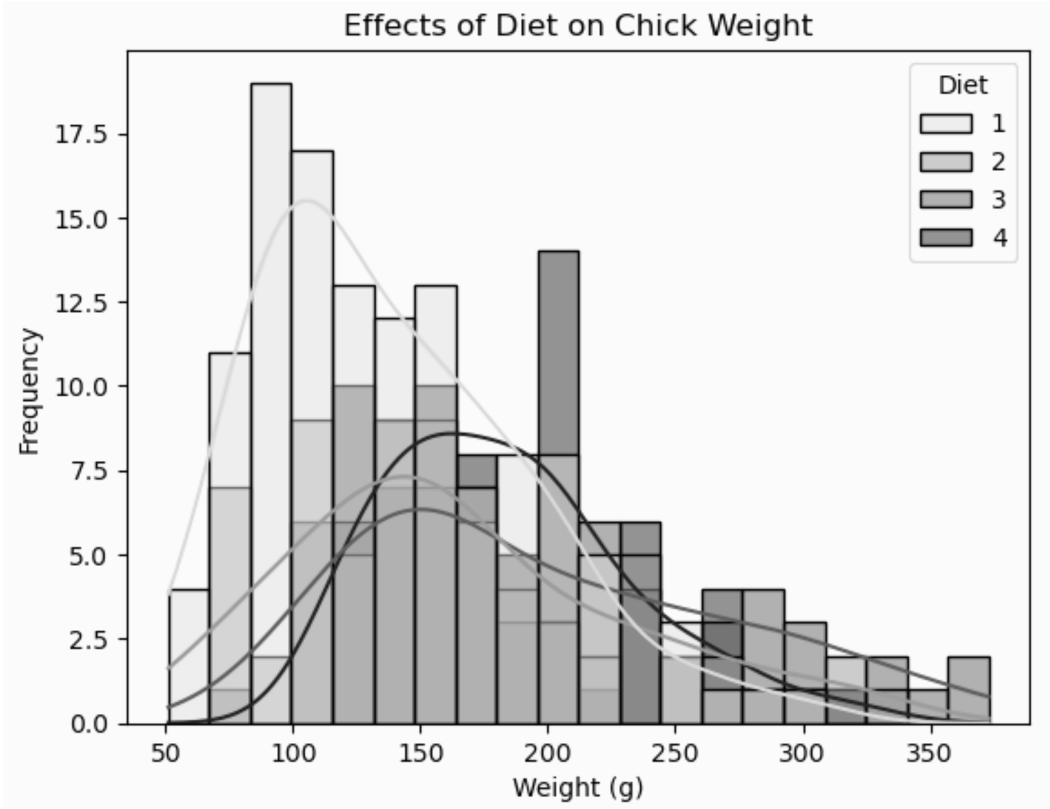
- Axis range
  - If you plot different panels (subplots) in one large figure, please care the value range and axis labels.



Notes from the previous assignment

- Color
  - Grayscale
  - Color-blind simulator





Notes from the previous assignment

- Color
  - Grayscale
  - Color-blind simulator

Sim Daltonism 4+

A window to color blindness

Michel Fortin

★★★★★ 4.9 • 430 Ratings

Free

View in Mac App Store ↗

Original

Deuteranopia

Deuteranomaly

Protanopia

Today's outline

- Plotting two variables
  1. Barplot
  2. Boxplot
  3. Jitter plot and beeswarm plot
  4. Violin plot

Plotting two variables

NUMERIC VARIABLE AND CATEGORICAL VARIABLE

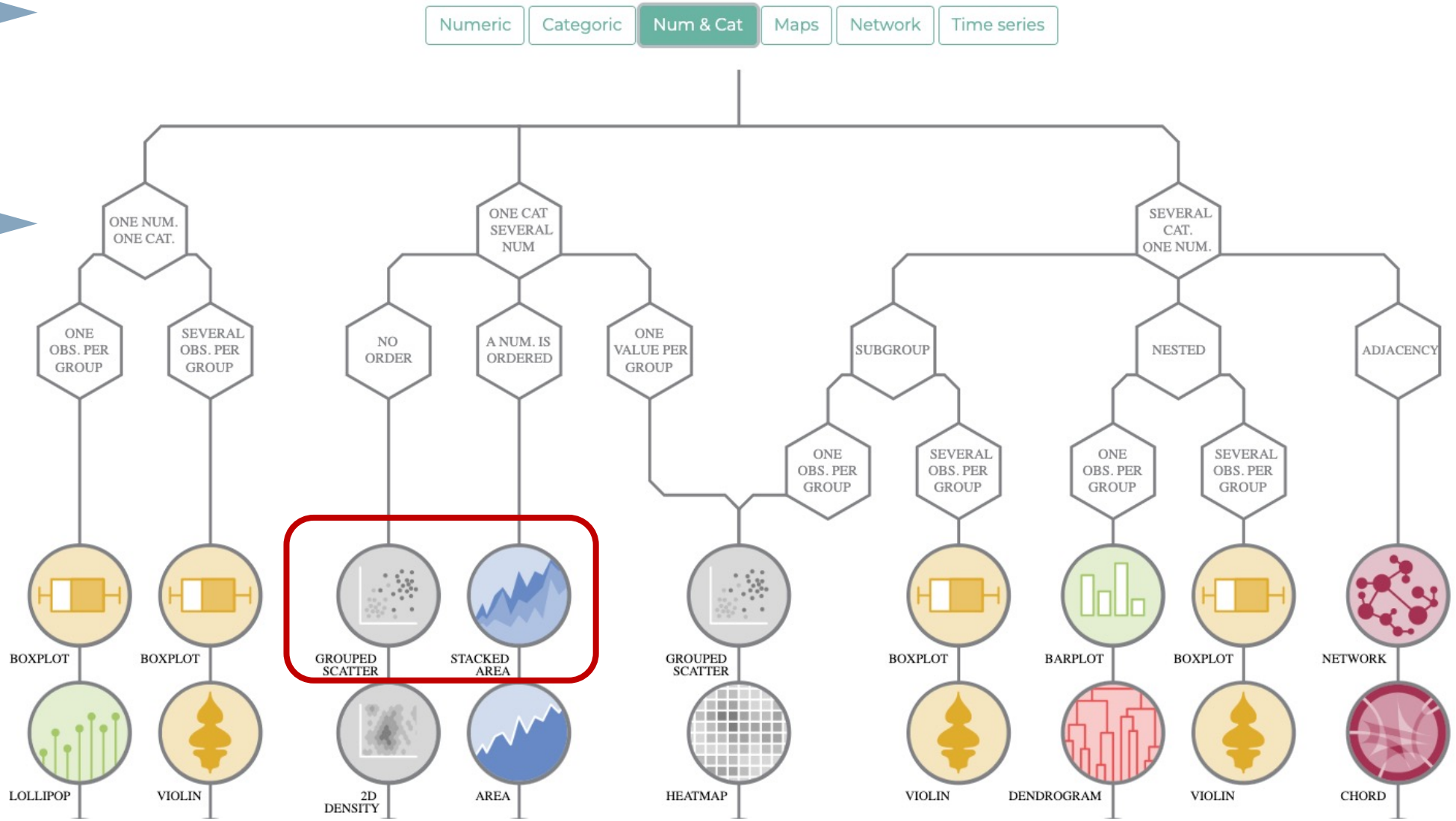
What should we consider when we visualize data?

1. The number of variable
2. Is the variable(s) numeric, or categorical?
   - **Two variables – Numeric & Categorical**
     1. Barplot
     2. Boxplot
     3. Jitter plot and beeswarm plot
     4. Violin plot

How many variables? Numeric or categorical?

Numeric or categorical?

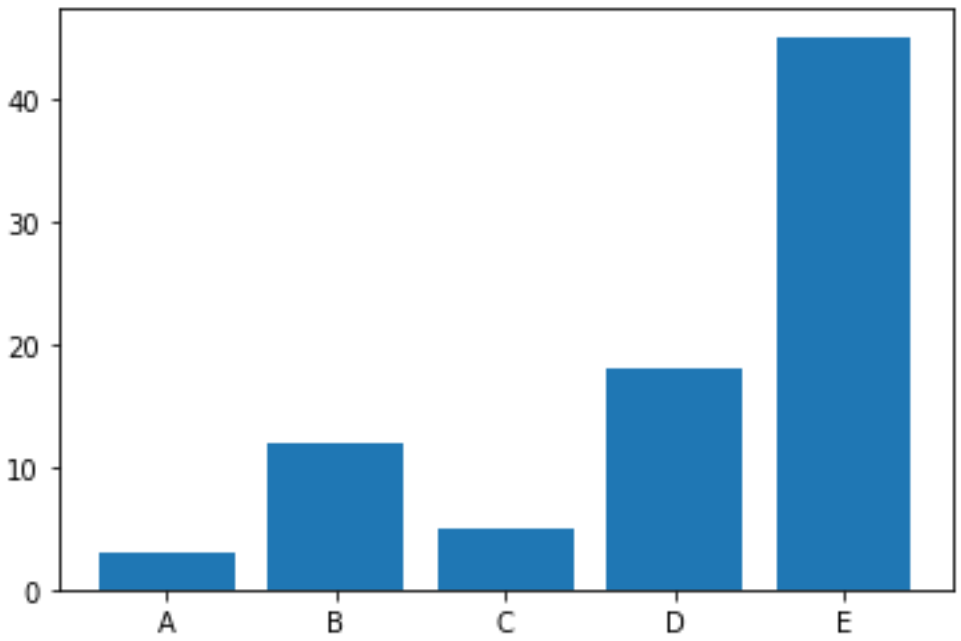
The number of variables



https://www.data-to-viz.com/

Barplot

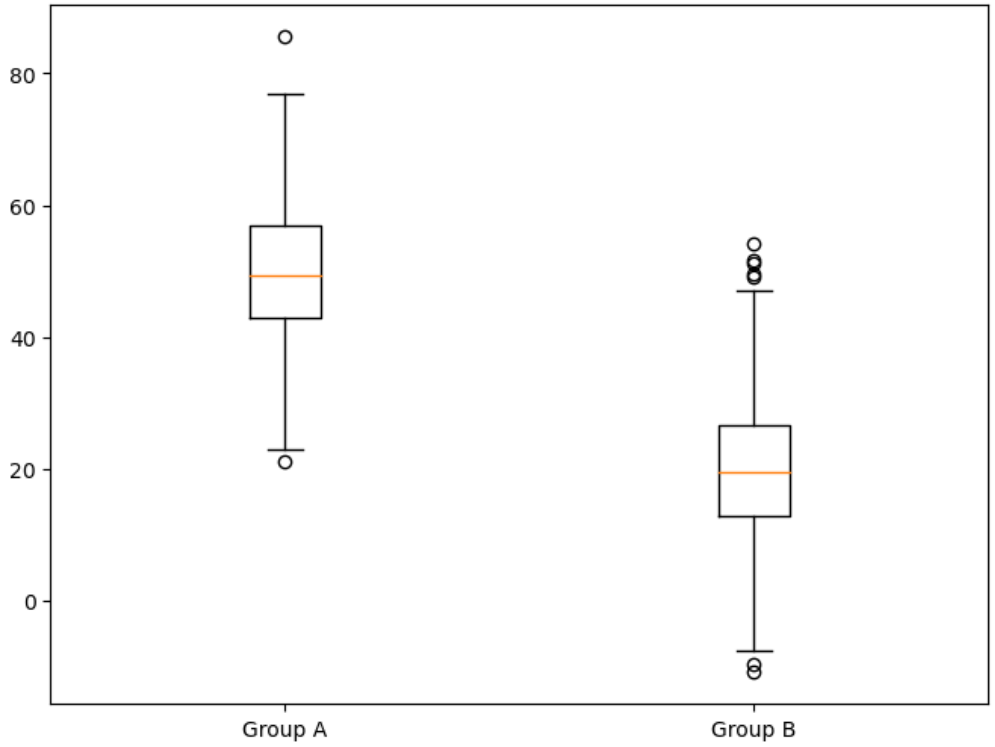
- A **barplot** shows the relationship between **a numeric and a categorical variable**.
- The **height** of the bar usually represents its numeric value.
- We often use error bars, but the simplest form represents only the **mean** or **median** of each data.



https://python-graph-gallery.com/barplot/

Boxplot

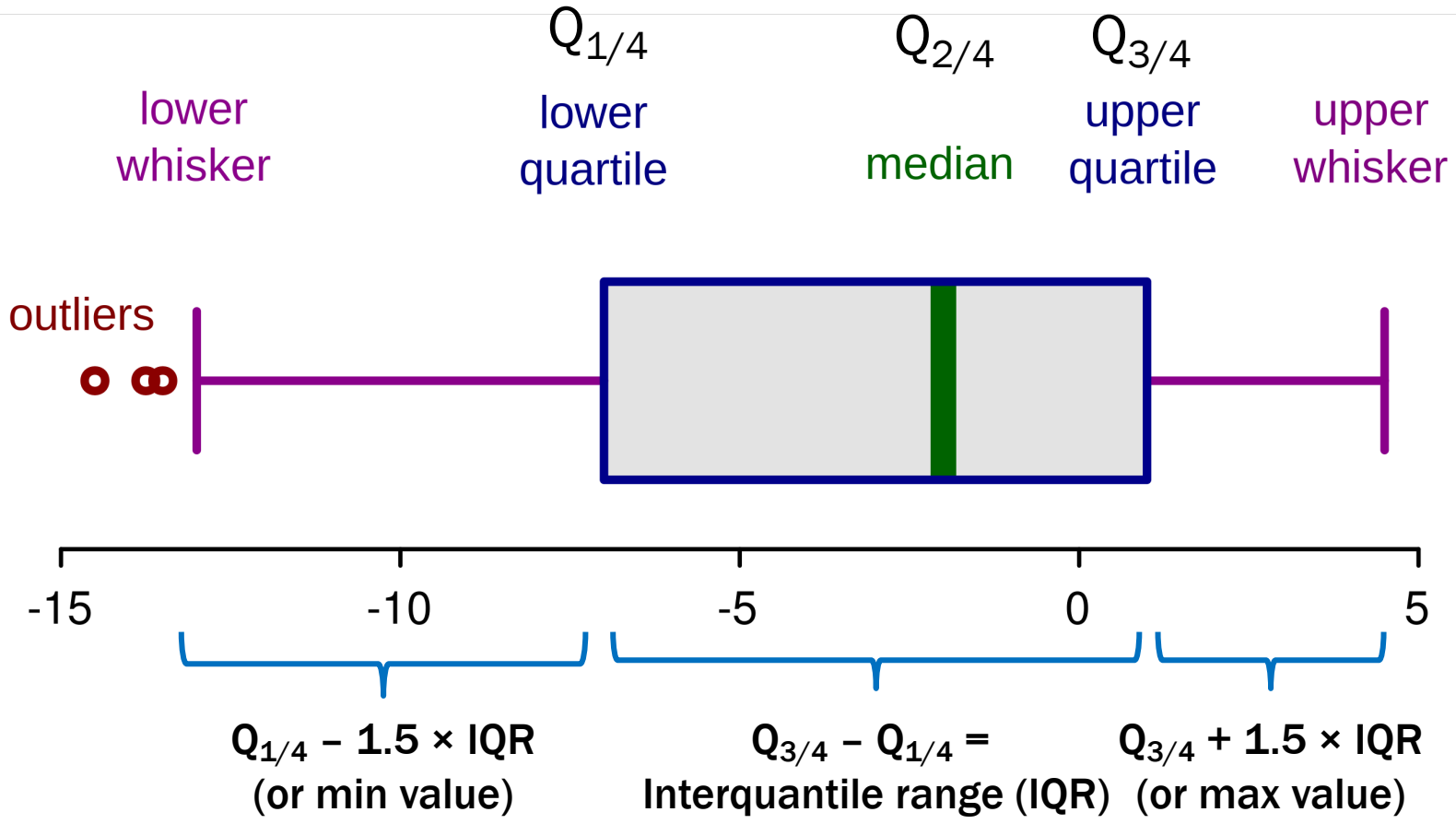
- A **boxplot** summarizes the distribution of **a numeric variable for one or several categories**.
- It allows to show the **median**, **quantiles**, and **outliers.**
- However, it **hides** detailed information about the original distribution.



https://python-graph-gallery.com/543-grouped-boxplots-matplotlib/

Boxplot

- $Q_{1/4}$ = 25% of data points are below this value
- $Q_{2/4}$ = 50% of data points are below this value
- $Q_{3/4}$ = 75% of data points are below this value



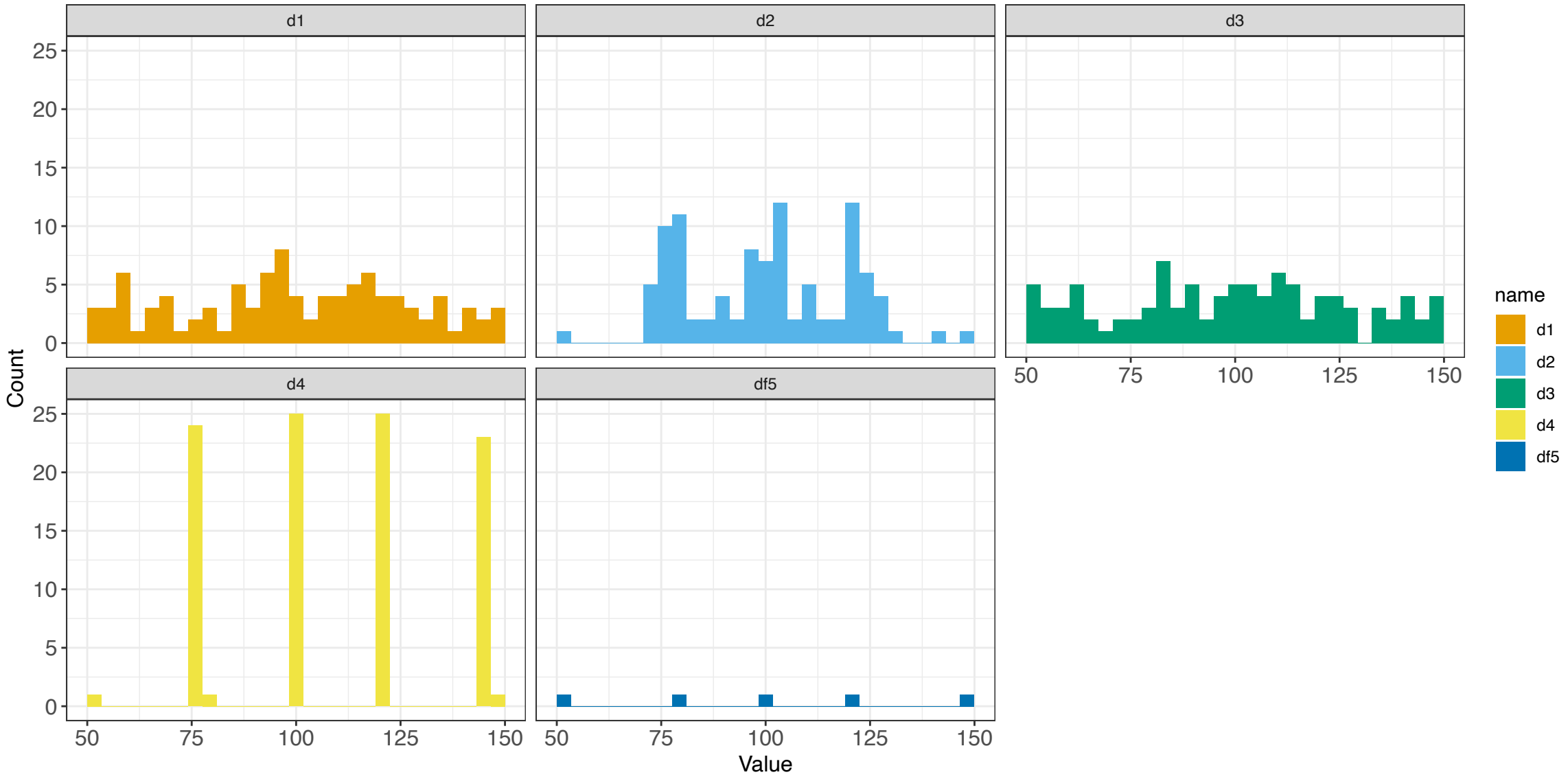
Ruediger85 (CC-BY-SA 3.0)

https://ja.wikipedia.org/wiki/%E7%AE%B1%E3%81%B2%E3%81%92%E5%9B%B3

Barplot and boxplot

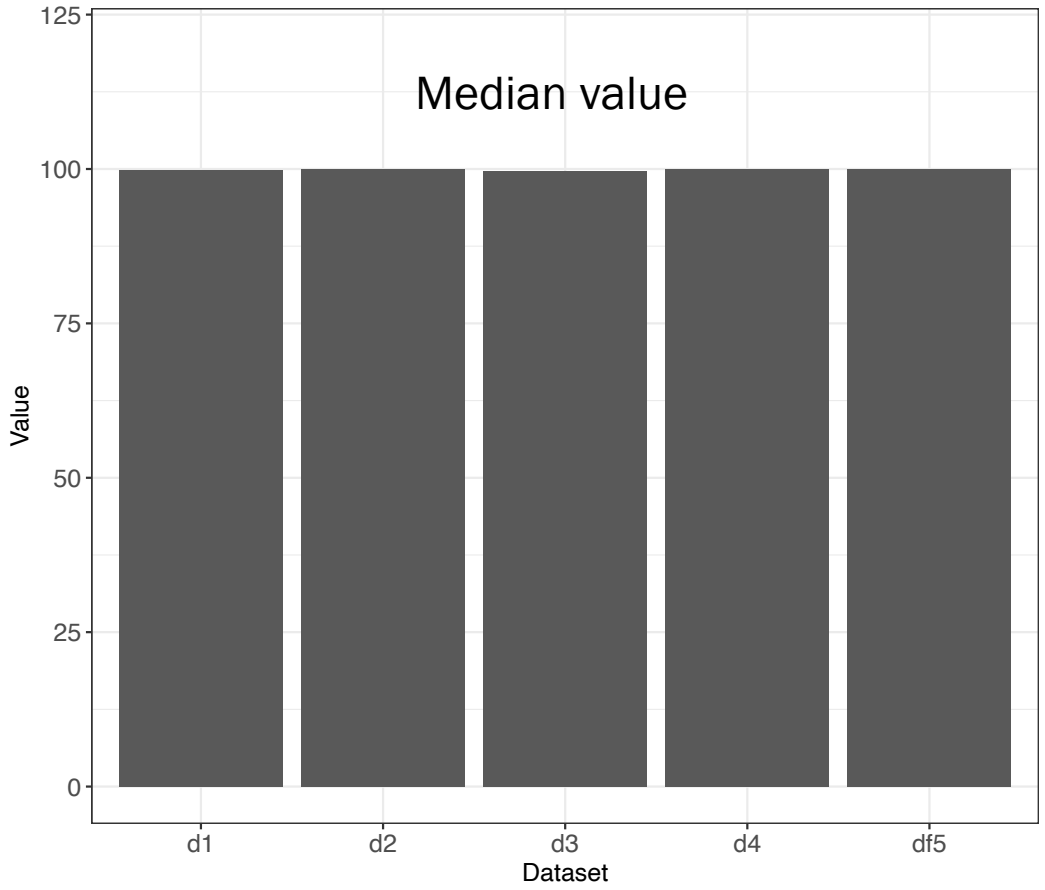
- **Barplot** and **boxplot** are easy to draw. They can be drawn even without a computer.
- But they can present **limited information only**. On barplot or boxplot, very different data look similar.

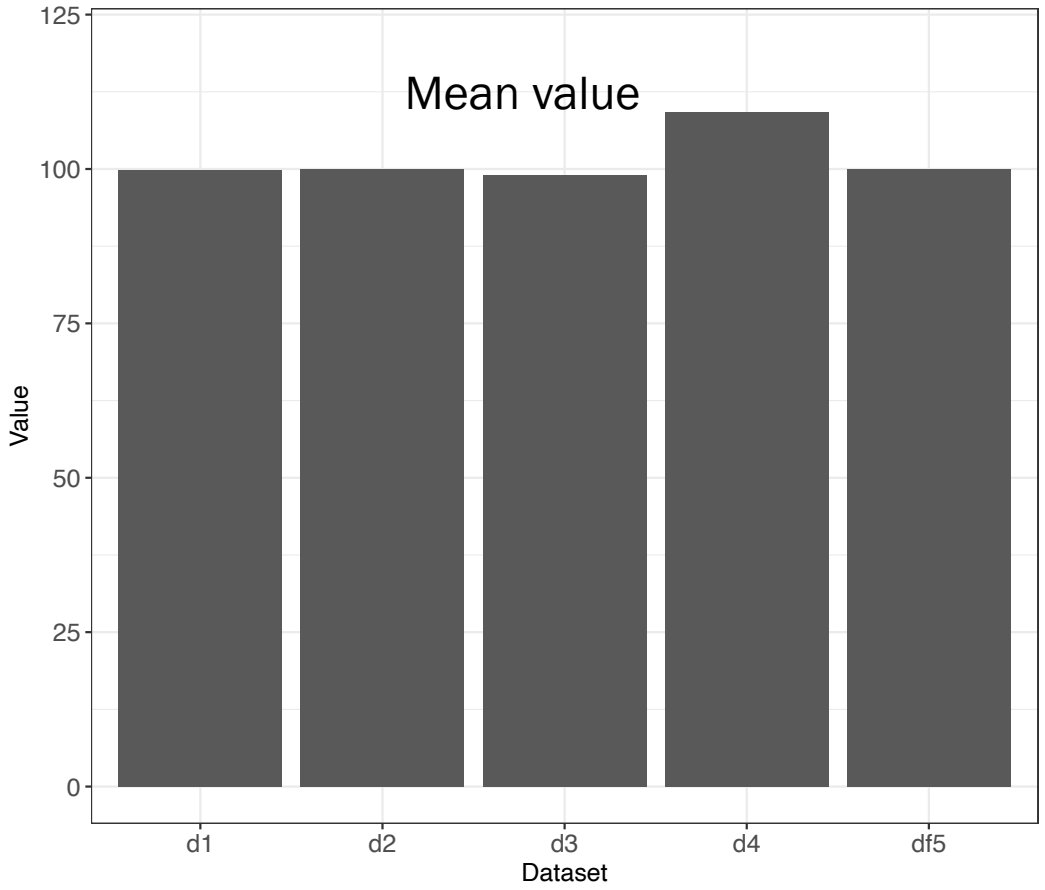
Barplot and boxplot



Barplot and boxplot

- I created five artificial datasets. **They are very different data**, but let's see how they look like if we use barplots and boxplots.





Barplot and boxplot

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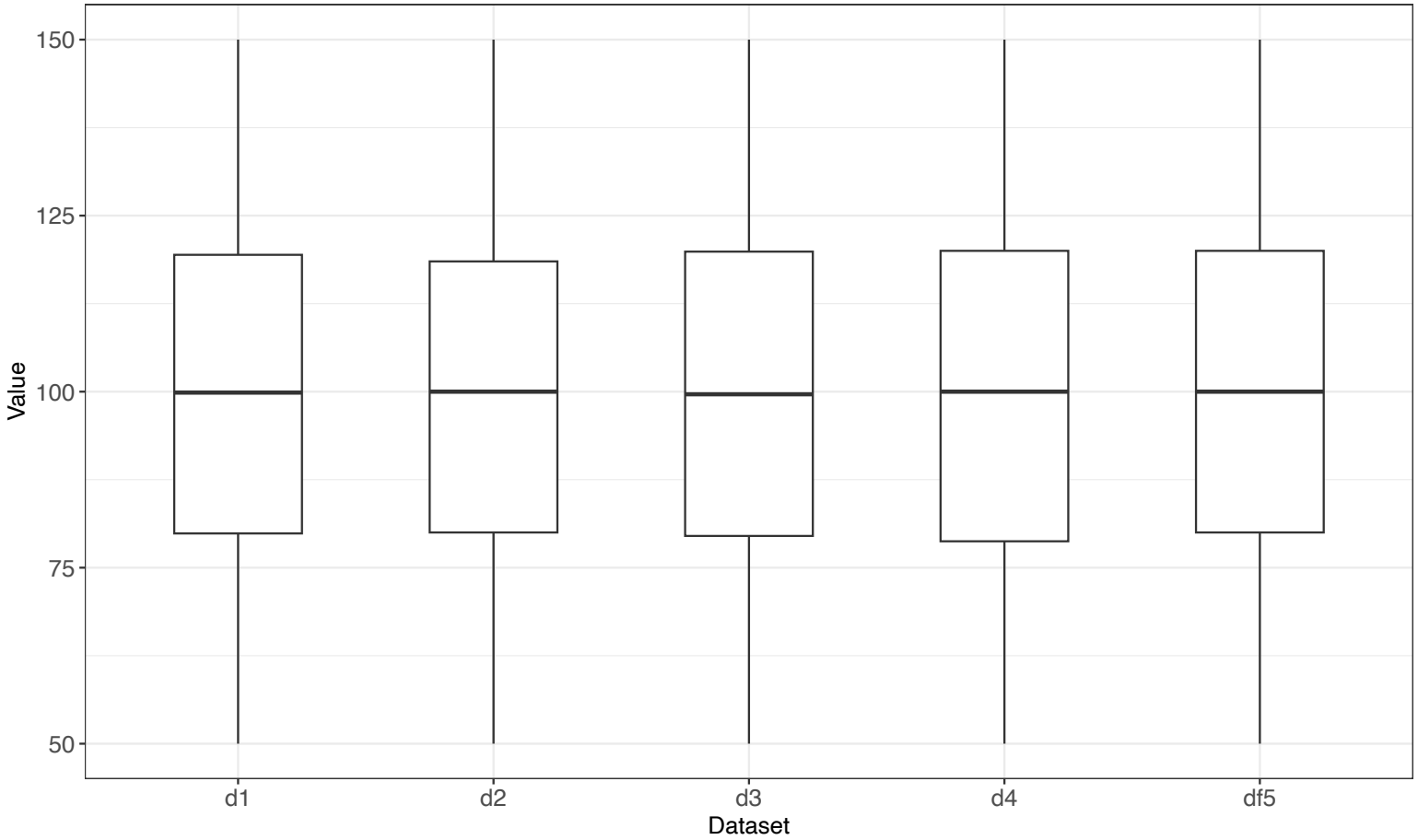
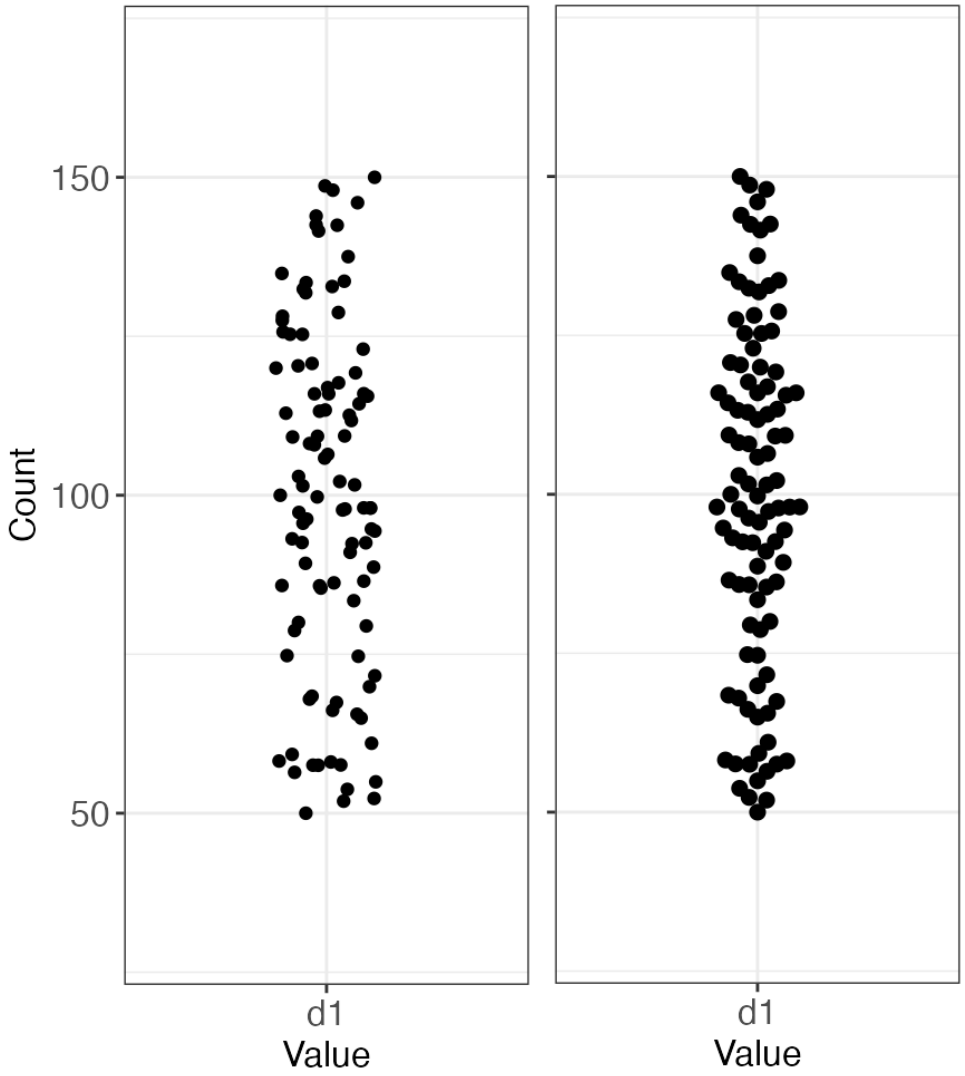


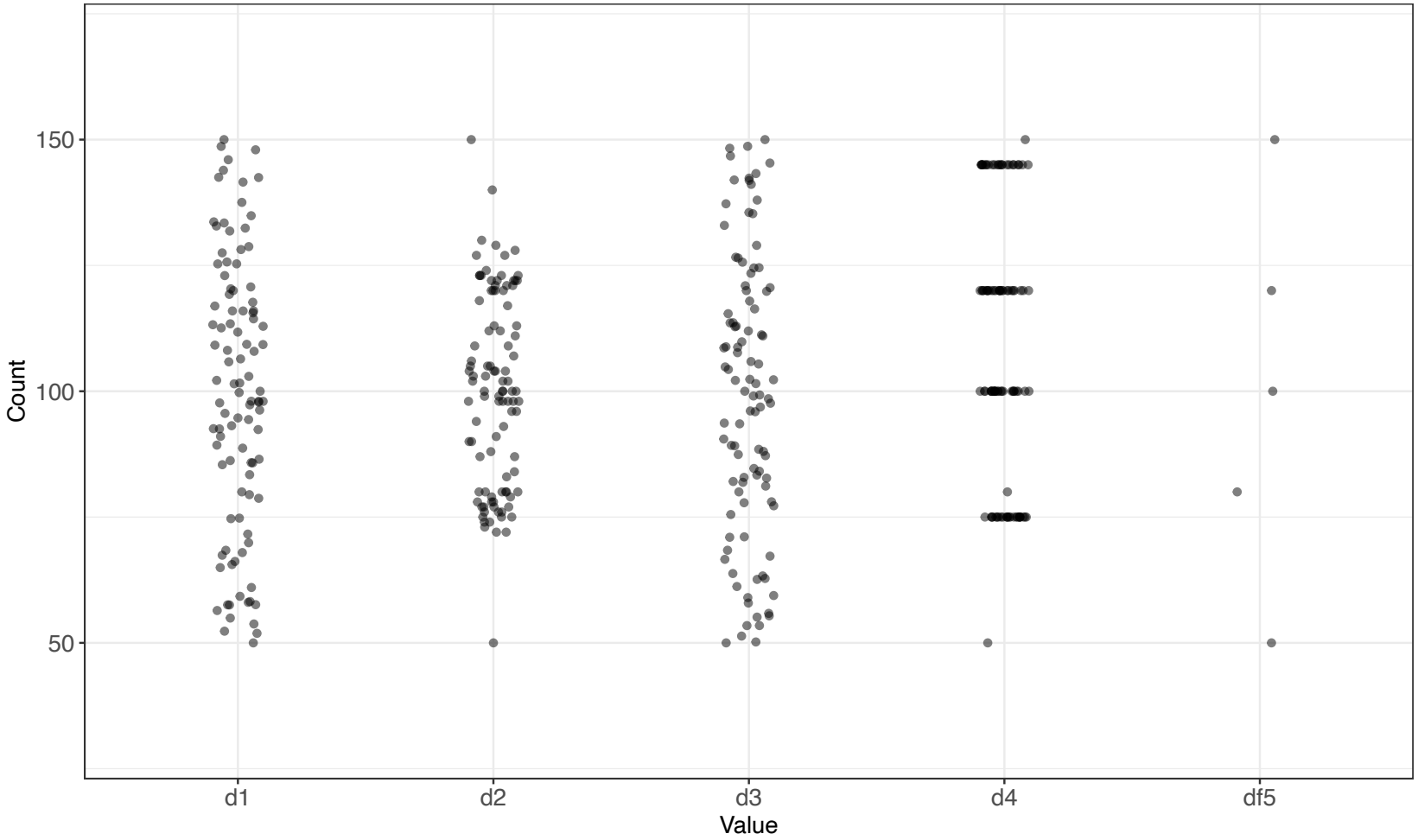
Jitter plot and beeswarm plot

- A **jitter plot** is used to visualize the distribution of many individual values.
- The values are plotted as points along one axis, and the dots are then shifted randomly along the other axis, allowing the dots not to overlap.
- A **beeswarm plot** (or swarm plot) is a similar plot, but the dots are shifted in a way that the dots cluster (but not overlap) towards its center.



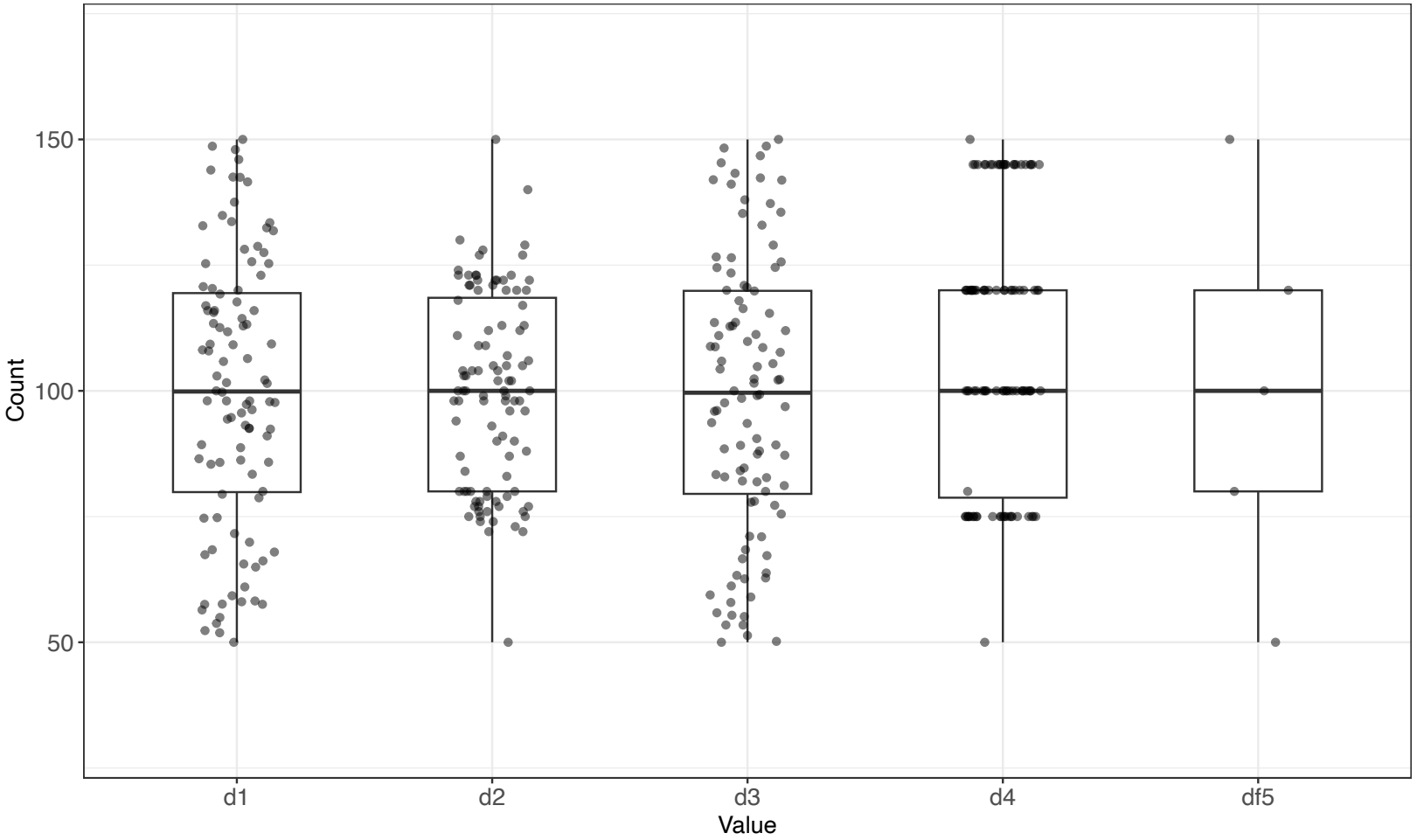
Jitter plot and beeswarm plot

- Let's plot the previous five datasets. Now the differences are clear.



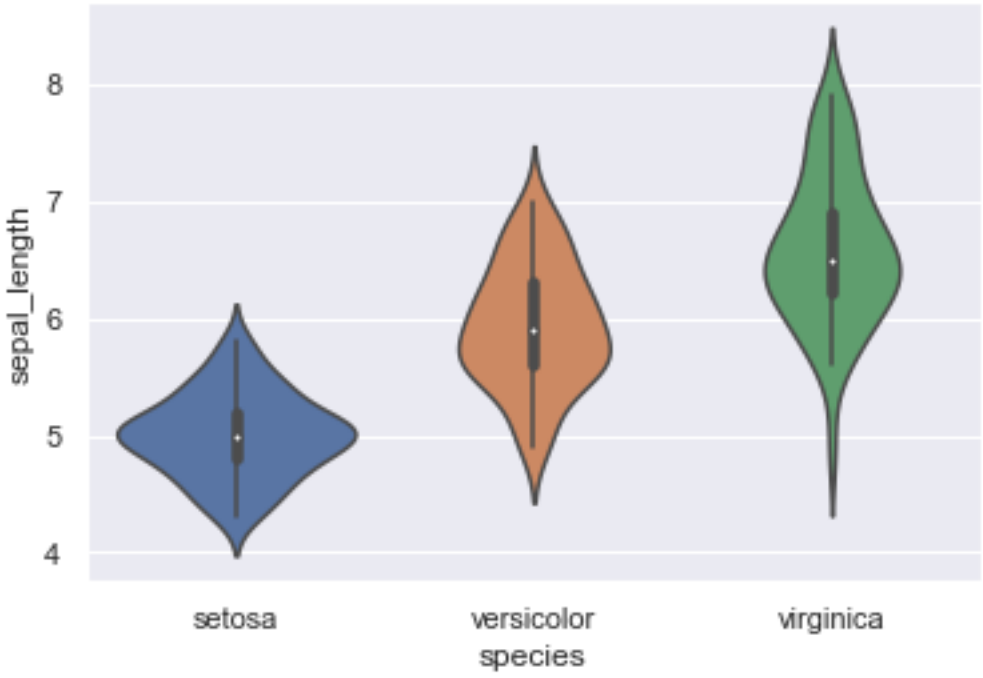
Jitter plot and beeswarm plot

- Combining boxplot and jitter plot is a good idea.



Violin plot

- A **violint plot** allows you to visualize the distribution of a numeric variable for one or several groups.
- The density distribution is plotted along one axis, and thus, it is intuitively easy to understand the distribution pattern.

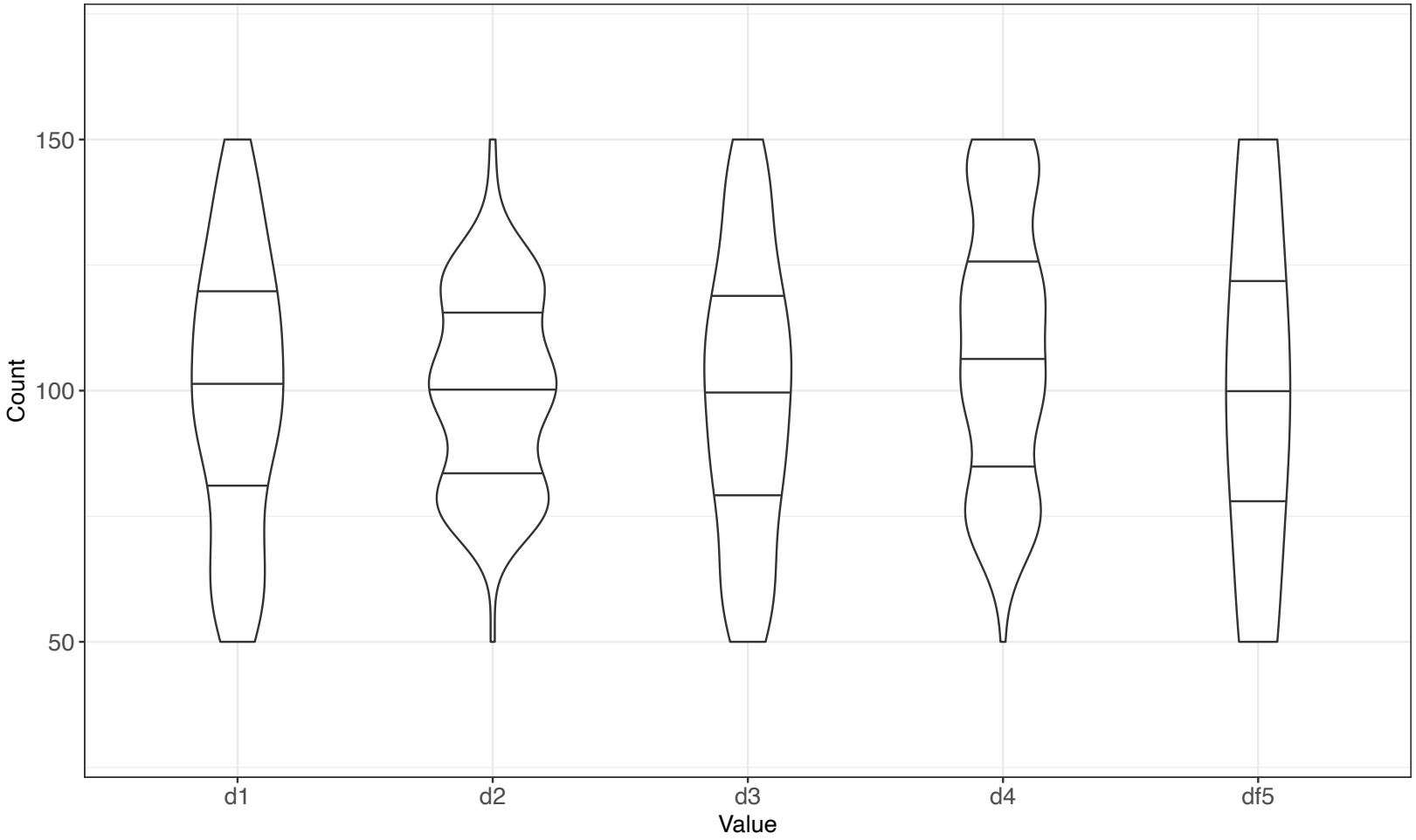


https://python-graph-gallery.com/50-basic-violinplot-and-input-formats/

Jitter plot and beeswarm plot

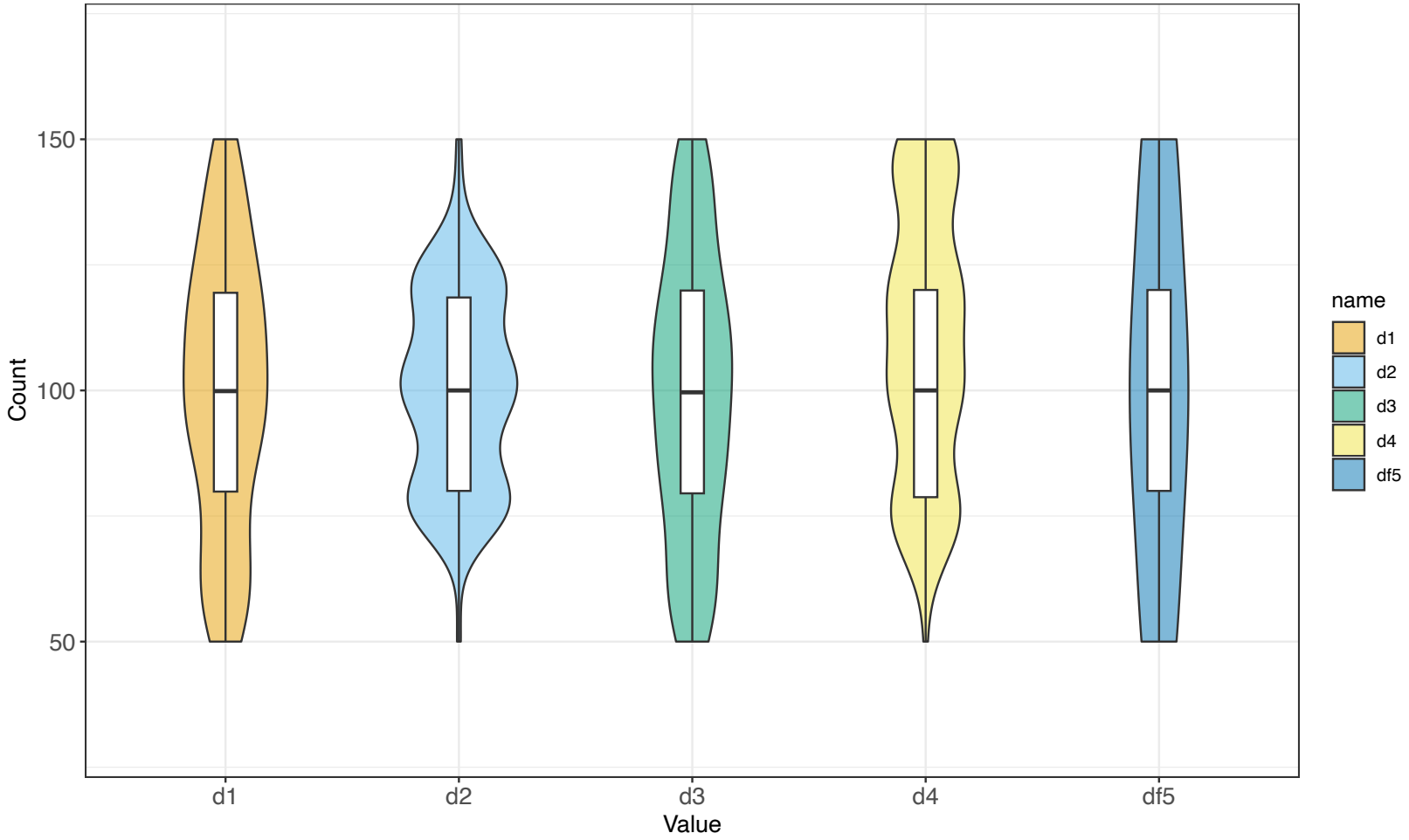
- Let's plot the previous five datasets. Now the differences are clear.

- $Q_{1/4}$
- $Q_{2/4}$
- $Q_{3/4}$



Jitter plot and beeswarm plot

- Combining boxplot and violin plot is also a good idea.



Mini summary

- **Barplot and boxplot**
  - Easy to draw (even without a computer)
  - Important information is missing
- **Jitter plot (and beeswarm plot)**
  - Visualize all individual data points
  - Probably suitable for medium-sized data
  - Used with boxplots is a good idea
- **Violin plot (and beeswarm plot)**
  - Visualize density distribution
  - Probably suitable for large-sized data
  - Used with boxplots is also a good idea